INCH-POUND

MIL-N-53098(ME)
7 February 1990

MILITARY SPECIFICATION

NOZZLES, FUEL AND OIL SERVICING, NONAUTOMATIC SHUTOFF,

ARCTIC SERVICE

This specification is approved for use within the USA Belvoir Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers nonautomatic shutoff nozzles used for dispensing liquid fuel and oil in all ambient temperatures between -60 and +95 °F.

1.2 Classification. The nozzles shall be the following sizes, as specified (see 6.2):

Size 1 - For use with 3/4-inch or 1-inch hose.
Size 2 - For use with 1-1/4-inch or 1-1/2-inch hose.
Size 3 - For use with 2-inch or 2-1/2-inch hose.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 specifications and standards. The following specifications and standards form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

A-A-1037	- Wire Fabric, Industrial.
QQ-P-416	- Plating, Cadmium (Electrodeposited).
PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-B-636	- Box, Shipping, Fiberboard.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 4930

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PPP-C-1752 - Cushioning Material, Packaging, Uncellular Polyethylene Foam, Flexible.
PPP-C-1842 - Cushioning Material, Plastic, Open Cell (for Packaging Applications).

MILITARY

MIL-P-116 - Preservation, Methods of.
MIL-G-3056 - Gasoline, Automotive, Combat.
MIL-T-5624 - Turbine Fuel, Aviation, Grades JP-4 and JP-5.
MIL-C-7024 - Calibrating Fluid, Aircraft Fuel System Components.
MIL-L-7808 - Lubricating Oil, Aircraft Turbine Engine, Synthetic Base.
MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-L-23699 - Lubricating Oil, Aircraft Turbine Engine, Synthetic Base.
MIL-C-81562 - Coatings, Cadmium, Tin-Cadmium and Zinc (Mechanically Deposited).
MIL-C-81740 - Coatings, Aluminum and Aluminum Alloys (Metallic Compound Decomposition).
MIL-C-81797 - Coating, Inorganically Bonded Aluminum (Electrophoretically Deposited).
MIL-T-83133 - Turbine Fuel, Aviation, Kerosene Type, Grade JP-8.
MIL-C-83413 - Connectors and Assemblies, Electrical, Aircraft Grounding.
MIL-C-83413/7 - Connectors and Assemblies, Electrical Aircraft Grounding: Grounding Clamp Convector for Types I and III Grounding Assemblies, Clip, Electrical.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.
FED-STD-595 - Colors.

MILITARY

MIL-STD-29 - Springs, Mechanical: Drawing Requirement for.
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-810 - Environmental Test Methods.
MIL-STD-870 - Cadmium Plating, Low Embrittlement, Electrodeposition.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MS25384 - Electrostatic Discharger Jumper, Fuel Nozzle-to-Aircraft.

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(Unless otherwise indicated, copies of federal and military specifications, standards are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government publications. The following other Government publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

CODE OF FEDERAL REGULATIONS (CFR)

Title 40, Part 80 - Regulation of Fuel and Fuel Additives.

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on subscription basis by the Superintendent of Documents, Government Printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B46.1 - Surface Texture.

(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 276 - Stainless and Heat-Resisting Steel Bars and Shapes.
A 313 - Chromium-Nickel Stainless and Heat-Resisting Steel Spring Wire.
A 380 - Cleaning and Descaling Stainless Steel Parts, Equipment and Systems.
A 525 - Steel Sheet, Zinc-Coated (Galvanized) by the Hot-Dip Process, General Requirements for.
A 582 - Free-Machining Stainless and Heat-Resisting Steel Bars, Hot-Rolled or Cold-Finished.
B 21 - Naval Brass Rod, Bar, and Shapes.
B 26 - Aluminum-Alloy Sand Castings.
B 36 - Brass Plate, Sheet, Strip, and Rolled Bar.
B 85 - Aluminum-Alloy Die Castings.
B 86 - Zinc-Alloy Die Castings.
B 108 - Aluminum-Alloy Permanent Mold Castings.
B 121 - Leaded Brass Plate, Sheet, Strip, and Rolled Bar.
B 124 - Copper and Copper Alloy Forging Rod, Bar, and Shapes.
B 135 - Seamless Brass Tube.

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B 159 - Phosphor Bronze Wire.
B 210 - Aluminum-Alloy Drawn Seamless Tubes.
B 271 - Copper-Base Alloy Centrifugal Castings.
B 505 - Copper-Base Alloy Continuous Castings.
B 584 - Copper Alloy Sand Castings for General Applications.
D 471 - Rubber Property - Effects of Liquids.
D 910 - Aviation Gasolines.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

UNDERWRITERS' LABORATORIES (UL)

UL 842 - Standard for Safety: Valves for Flammable Fluids.

(Application for copies should be addressed to the Underwriters' Laboratories, Inc., 207 East Ohio Street, Chicago, IL 60611; 333 Pfingsten Road, Northbrook, IL 60062; 1283 Walt Whitman Road, Melville, LI, NY 11746; 1655 Scott Boulevard, Santa Clara, CA 95050; or 2602 Tampa East Boulevard, Tampa, FL 33619.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, (except for related associated detail specifications, specification sheets or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The nozzles as shown in figure 1, intended for attachment to fuel hoses, shall consist of a spout, a control device as specified herein, a body, and a carrying handle to accommodate an arctic mitten (see 3.4.3).

3.2 First article. Unless otherwise specified specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor, shall be subject to all provisions of this specification, and shall be of the quality specified in UL 842 for approved fuel servicing nozzles, resistant to corrosion in all grades of gasoline and jet hydrocarbon fuels.

3.3.1 Material deterioration prevention and control. The nozzles shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environments to which the nozzles may be exposed.

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3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the nozzles may be newly fabricated from recovered materials to the maximum extent practicable, provided the nozzles produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the nozzles.

3.3.3 Body. The nozzle body shall be cast aluminum alloy conforming to ASTM B 26, alloy UNS A03560, temper T6; alloy UNS A07120, temper T5; alloy UNS A05350, temper F; of ASTM B 85, alloy UNS A03800.

3.3.4 Valve springs. The valve springs shall be copper wire conforming to ASTM B 159, alloy UNS C51000; corrosion-resisting steel wire conforming to ASTM A 313, alloys UNS S30200, S30400, or S31600; or spring steel plated in accordance with QQ-P-416, class 1, type II. In addition, the spring steel can be coated according to MIL-STD-870, class 1, type I; or aluminum coated in accordance with MIL-C-81740, class 1, type II, or equivalent; or coated in accordance with MIL-C-81562, class 1, type I; or a metallic ceramic coating (0.001-inch thickness) in accordance with MIL-C-81797.

3.3.5 Valve stem. The valve stem shall be corrosion-resisting steel conforming to ASTM A 276, alloys UNS S30200, S30400, or S31600, or ASTM A 582, UNS S30300, condition optional, cold finish and passivated in accordance with ASTM A 380.

3.3.6 Packing. The valve stem packing shall be a solid, machined or preformed tetrafluoroethylene polymer or a low temperature elastomer.

3.3.7 Spouts. The spouts shall be style 1, curved, or style 2, straight, as specified herein, and shall be seamless brass tubing conforming to ASTM B 135, alloy UNS C33000; or aluminum tube conforming to ASTM B 210, alloy UNS A96061, temper T6 or T4 or alloy A96063, temper T832.

3.3.8 Bushing, reducing. Unless otherwise specified (see 6.2), the reducing bushing shall be brass bar conforming to ASTM B 21, B 36, B 121, or B 124, alloys UNS C26000, C26800, C27200, C33500, C34000, C34200, C35000, C35300, C35600, C37700, C46200, C46400, C48200, C48500 (where appropriate); or brass castings conforming to ASTM B 271, B 505, or B 584, alloys UNS C83600, C83800, C84200, C84400, C84800 (where appropriate). Aluminum bushings shall be in accordance with ASTM B 26 or ASTM B 108, alloy UNS A03560, temper T6. Zinc bushings shall be in accordance with ASTM B 86, alloy UNS Z33520.

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3.3.9 Check valve. Unless otherwise specified (see 6.2), a check valve shall not be required. When a check valve is required, it shall be compatible with the nozzle and shall be in accordance with the manufacturer's standard check valve.

3.4 Construction. The nozzles shall meet the performance requirements specified in 3.5, and shall be constructed to withstand the strains, jars, vibrations, and other conditions incidental to shipping, storage, and service usage. Except as otherwise specified herein, the nozzle shall conform to the assembly and design requirements of UL 842.

3.4.1 Weight. The size 1 nozzle shall weigh not more than 4.5 pounds. The size 2 nozzle shall weigh not more that 5.5 pounds. The size 3 nozzle shall weigh not more than 6.0 pounds. The nozzle weight shall not include the accessories specified herein.

3.4.2 Flow control. The operator shall initiate and maintain flow through the nozzle valve while utilizing a minimum operator activated force. As shown in figure 1, engagement and operation of the nozzle valve shall not be by squeezing an operating lever. The nozzle flow shall not be through a handle. Hold-open notches, lugs or lever supports shall not be used.

3.4.3 Nozzle arctic operation. The nozzle shall allow for handling and operation using an arctic mitten. Any handle that shall be used with an arctic mitten shall require a minimum hand width of 5.25 inches and finger height of 3.0 inches when gripping the handle.

3.4.4 Threads. All pipe and machine screw-threads shall conform to the applicable requirements of FED-STD-H28.

3.4.5 Valve stem surface. The valve stem surface that shall come in contact with the packing during operation of the nozzle shall have a surface roughness not greater than 16 microinches in accordance with ANSI B 46.1.

3.4.6 Interchangeability. All parts of the nozzle shall be functionally and dimensionally interchangeable (see 6.5).

3.4.7 Inlet.

3.4.7.1 Size 1. The inlet of the size 1 nozzle shall have internal nominal 1.0-inch American National Standard taper pipe thread (NPT).

3.4.7.2 Size 2. The inlet of the size 2 nozzle shall have internal nominal 1.5-inch NPT.

3.4.7.3 Size 3. The inlet of the size 3 nozzle shall have internal nominal 2.0-inch NPT.

3.4.8 Spout. The spout shall be style 1, nonflexible, curved type, or style 2, straight type, conforming to the contractor's standard configuration and length, provided the end-to-end length of the spout is not less than 7.0 inches

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prevent loss of the dust cap. The tapped hole in the nozzle handle

as measured along the centerline of the spout. The spout of the size 1 nozzle shall meet the release or breakaway requirements of UL 842.

3.4.8.1 Spouts for use with gasoline. Gasoline-dispensing nozzles shall be for use with leaded or unleaded gasoline, as specified (see 1.2), in accordance with CFR, title 40, part 80, as specified herein.

3.4.8.1.1 Class A, leaded-gasoline spout. The terminal end of the nozzle spout for use with leaded gasoline shall have an outside diameter of not less than 0.930 inches and a wall thickness of not less than 0.065 inches.

3.4.8.1.2 Class B, unleaded-gasoline spout. The terminal end of the nozzle spout for use with unleaded gasoline shall have an outside diameter of not more than 0.840 inches and a wall thickness of not less than 0.065 inches. The delivery end of the spout shall include a straight section not less than 2.50 inches in length. The retaining spring shall terminate 3.0 inches from the terminal end of the spout.

3.4.9 Accessories.

3.4.9.1 Bushing, reducing. The size 1 nozzle shall be furnished with an external nominal 1.0-inch x 0.75-inch NPT reducing bushing. The size 2 nozzle shall be furnished with an external nominal 1.50-inch x 1.25-inch NPT reducing bushing. The size 3 nozzle shall be furnished with an external nominal 2.0-inch x 1.5-inch NPT reducing bushing.

3.4.9.2 Dust cap, chain, and spring. An aluminum dust cap shall be attached to the nozzle by a chain and a spring. The cap, chain, and spring assembly shall withstand the pull test specified herein.

3.4.9.2.1 Chain. The chain shall be size no. 8 sash chain and shall be coated steel according to MIL-STD-870, class 1, type I; or aluminum coated in accordance with MIL-C-81740, class 1, type II, or equivalent; or coated in accordance with MIL-C-81562, class 1, type I; or a metallic ceramic coating (0.001-inch thickness) in accordance with MIL-C-81797, galvanized steel as per ASTM A 525 to a G90-coating designation, brass, or aluminum, and shall be attached to the nozzle by an S-hook. The chain shall have a working tensile strength of not less than 70 pounds. Dissimilar materials, metals or coatings used must be in conformance with MIL-STD-889.

3.4.9.2.2 Spring. The chain shall be tensioned by a 3-pound per inch, ±1-pound per inch, compression spring of phosphor bronze or stainless steel used as an extension spring to produce a yokelike drawbar action in accordance with MIL-STD-29. The chain shall not break and the spring shall not be extended beyond its original length when the load is removed, but shall allow the cap to be snapped in place or removed from inlet and spout tip.

3.4.9.3 Ground cable assembly. The nozzle assembly shall be equipped with a ground cable assembly for grounding the nozzle to the vehicle and to a ground stake. The nozzle-to-vehicle ground cable shall conform to MS25384-1, except the connector shall conform to MIL-C-83413/7. The nozzle-to-ground stake cable shall be coated wire 15 feet long, and the end shall be equipped with a clamp in

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accordance with MIL-C-83413/7, suitable for bonding the cable assembly to a 5/8-inch diameter ground stake.

3.4.9.4 Strainer. The spout tube of the size 2 and size 3 nozzles shall be equipped with a wire-cloth strainer. The wire cloth shall be 100 x 100 mesh (open area 30.3 percent), made from 0.0045-inch diameter round wire in accordance with A-A-1037, type I, class 1, grade A, from one of the compositions shown below:

a. Corrosion-resistant steel, 300 series, plain commercial finish.
b. Brass low-zinc content (80 percent copper and 20 percent zinc).
c. Copper-zinc alloy; commercial bronze (90 percent copper and 10 percent zinc).

The strainer and spout shall be capable of being inserted in, or removed from, the nozzle assembly by hand without the use of any tool. The nozzle spout and strainer shall be securely mounted to the nozzle to prevent any movement of the strainer or spout when assembled. The strainer-spout assembly shall meet the leak test performance specified herein.

3.4.10 Human factors engineering. The nozzles shall conform to human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given but not limited to paragraphs 4 (General Requirements), 5.4 (Controls), 5.9 (Design for Maintainability), and 5.13 (Hazards and Safety) of MIL-STD-1472 as applicable.

3.5 Performance characteristics.

3.5.1 Hydrostatic strength. All parts of the nozzle which are subjected to normal operation pressure shall be capable of withstanding, without rupture or deformation, hydrostatic pressure of not less than 250 pounds per square inch (psi) when tested as specified herein.

3.5.2 Leakage. There shall be no evidence of leakage past the valve seat, through the body, or through any of the joints or seals of the nozzle when tested as specified herein.

3.5.3 Mechanical strength. The nozzle shall withstand, without evidence of a malfunction or leakage, the mechanical strength test specified herein.

3.5.4 Nozzle capacity. The nozzles shall have capacities of not less than the flows specified in table I and shall be tested as specified herein.

TABLE I. Capacity (gasoline).

Size	Nozzle inlet pressure (max) (psi)	Capacity (gpm)
1	15	12
2	15	40
3	10.2	136

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3.6 Marking for identification. Each nozzle shall be permanently and legibly marked with the manufacturer's identification in accordance with MIL-STD-130.

3.7 Anodizing finish. All aluminum and aluminum alloy components of the nozzle shall be anodized in accordance with MIL-A-8625, type II, class 2. The anodic coating of visible components shall be dyed green, color no. 34094 of FED-STD-595.

3.8 Fuel resistance. The nozzle shall be resistant to, and suitable for use with, fuels and fluids conforming to MIL-G-3056, ASTM D 910, MIL-T-5624, MIL-C-7024, MIL-L-7808, MIL-L-23699, MIL-T-83133, and ASTM D 471, reference fuel B.

3.9 Environmental.

3.9.1 Operating temperature. The nozzle shall perform as specified herein at all ambient temperatures between -60 °F and +95 °F.

3.9.2 Storage temperature. The nozzle shall not be damaged by storage at all ambient temperatures between -65 °F and +145 °F.

3.10 Workmanship. The nozzle shall be internally and externally clean of deleterious ingredients and shall be free of sharp edges, burrs, cracks, chips, and gouges. All parts, components, and assemblies of the nozzle which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

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4.1.2 Component and material inspection. The contractor is responsible for insuring that components and materials are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).
d. Inspection comparison (see 4.7).

4.3 First article inspection. Unless otherwise specified (see 6.2), two first article nozzles shall be examined and tested as specified in 4.5.1 and 4.5.2. Presence of one or more defects and failure of any test shall be cause for rejection. Unless otherwise specified (see 6.2), the Government reserves the right to perform first article testing on three or more nozzles. The tests shall be those specified in 4.5.2 which the Government desires to perform.

4.3.1 Rejection and retest. When a nozzle fails to meet any of the requirements of the first article tests, the cause of failure shall be determined. Full particulars of the failure and the action required to correct the fault shall be reported to the contracting officer. All first article tests which are affected by the corrections shall be repeated.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105. Sample size shall be determined using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.4.2 Examination. Samples selected in accordance with 4.4.1 shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.4.3 Tests. Samples selected in accordance with 4.4.1 shall be tested in accordance with, and shall conform to the requirements of, UL 842, and as specified in 4.5.2.2, 4.5.2.2.1, 4.5.2.2.2, 4.5.2.4, and 4.5.2.5. Failure of any test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The nozzle shall be examined as specified herein for the following defects:

101. Materials not as specified.
102. Materials not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration in the applicable storage and operating environments.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.

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104. Contractor does not have documentation available for identification of materials, material finishes, or treatment.
105. Used, rebuilt, or remanufactured components, pieces, or parts incorporated in the nozzles.
106. Nozzle assemblies not as specified.
107. Parts missing or not as specified.
108. Construction not as specified.
109. Weight not as specified.
110. Controls not as specified.
111. Dimensions not as specified.
112. Threads not as specified.
113. Interchangeability of like parts not as specified.
114. Any accessory missing or not as specified.
115. Functioning component inoperative or not functioning as specified.
116. Human factors engineering not as specified.
117. Identification marking missing, incomplete, or illegible.
118. Finish not as specified.
119. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Unless otherwise specified herein, tests shall be performed in any ambient temperature between -60 °F and +95 °F.

4.5.2.2 Hydrostatic strength. With the valve seat in the closed position, the nozzle shall be subjected to an internal hydrostatic pressure employing MIL-C-7024, type II, Stoddard's solvent solution for first article nozzles and production nozzles, at the same ambient temperature. The pressure shall be raised slowly in 30-psi increments to not less than 250 psi and then held at the pressure for not less than one minute. Nonconformance to 3.5.1 shall constitute failure of this test.

4.5.2.2.1 Leakage with valve, closed, outlet open. With the valve in the closed position, the outlet open, and the nozzle immersed in water, the nozzle poppet shall be subjected to not less than 2 psi air pressure applied at the nozzle inlet. The air pressure shall then be increased to not less than 30 psi and the nozzle examined for body and poppet leakage. Nonconformance to 3.5.2 shall constitute failure of this test.

4.5.2.2.2 Leakage with valve open, outlet closed. With the valve in the open position, the outlet closed, and the nozzle immersed in water, not less than 5 psi air pressure shall be applied throughout the nozzle. Nonconformance to 3.5.2 shall constitute failure of this test.

4.5.2.3 Mechanical strength. The size 1 nozzle shall be tested for mechanical strength in accordance with UL 842. The size 2 and size 3 nozzles shall undergo the same mechanical strength test as the size 1 nozzle, except that size 2 and size 3 nozzles shall be attached to not less than 10 feet of 1.250-inch gasoline hose. Nonconformance to 3.5.3 shall constitute failure of this test.

4.5.2.4 Nozzle flow test.

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4.5.2.4.1 Flow capacity. The nozzle shall be tested at an ambient temperature of +95 °F, by discharging Stoddard's solvent solution at the same ambient temperature. The capacity of the nozzles shall be determined under pressure when measured at a distance of not less than 10 pipe diameters upstream of the nozzle inlet. Not less than 500 gallons shall be discharged through the nozzle when determining the flow in gallons per minute. Repeat both of the above procedures at an ambient temperature of -60 °F. Inability of the nozzle to shut off under any of the above test conditions shall constitute failure of this test. Any leakage of the nozzle shall constitute failure of this test. All flow rate performance data shall be corrected to a specific gravity of 0.78 (see 4.5.2.4.2).

4.5.2.4.2 Performance. The nozzles shall be installed in a test setup which will provide a regulated maximum inlet pressure of 15 ±2 psi (sizes 1 and 2) and 10 ±2 psi (size 3) under all flow conditions from no flow to the maximum flow rate. The nozzle valve shall be cycled open and closed at a rate not faster than 6 cycles per minute for 300 cycles at -60 °F, 150 cycles at -50 °F, 150 cycles at -40 °F, 150 cycles at -30 °F, 150 cycles at -10 °F, 300 cycles at +10 °F, 300 cycles at +50 °F and 300 cycles at +95 °F. The test shall include an equal number of cycles for size 1 at maximum (12 gpm) and low (approximately 6 gpm) flow rates; for size 2 at maximum (40 gpm), intermediate (20 gpm), and low (approximately 10 gpm) flow rates; for size 3 at maximum (136 gpm), intermediate (70 gpm), and low (approximately 30 gpm) flow rates. No maintenance shall be performed on the nozzles during this test. Any evidence of external leakage, cracking, sticking of the valve, or operating handle becoming inoperative, shall constitute failure of this test.

4.5.2.4.3 Low temperature. The nozzle shall be tested in accordance with MIL-STD-810, method 502.2, procedures I (storage), II (operation), and III (manipulation). Immerse the nozzle in test fluid conforming to ASTM D 471, reference fuel B at a temperature of -65 °F maximum for a period of 48 hours. After the 48-hour soak period, and with the ambient and test fluids at a temperature no higher than -60 °F, test the nozzle as specified in 4.5.2.4.1 and 4.5.2.4.2. Non-conformance to 3.4.3, 3.5.4 and 3.9 shall constitute failure of this test.

4.5.2.4.4 High temperature. The nozzle shall be tested in accordance with MIL-STD-810, method 501.2, procedures I (storage) and II (operation). Immerse the nozzle in test fluid conforming to ASTM D 471, referrence fluid B at a temperature of +95 °F for a period of 48 hours. After the 48-hour soak period, and with the ambient test fluids at a temperature no higher than +95 °F, test the nozzle as specified in 4.5.2.4.1 and 4.5.2.4.2. Nonconformance to 3.4.3, 3.5.4 and 3.9 shall constitute failure of this test.

4.5.2.4.5 Fuel resistance. The tests specified in 4.5.2.4.5.1 through 4.5.2.4.5.4 shall be conducted in the sequence listed and each test shall follow immediately after the preceding test.

4.5.2.4.5.1 Fuel soak, phase I. Immerse the nozzle in test fluid conforming to ASTM D 471, reference fuel B at a temperature of +95 °F for a period of 14 days. During the first 7 days, the control valve shall be in the closed

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normal functioning of the nozzle. ... interfere wi

position. During the last 7 days, the flow control valve shall be in the open position. After a minimum of 14 days fuel soak, the nozzle shall be tested as specified in 4.5.2.4.2.

4.5.2.4.5.2 Drying cycle, phase I. Drain all test fluid from the nozzle and allow it to dry for 24 hours in an ambient air temperature of +95 °F. The nozzle then shall be tested as specified in 4.5.2.4.2.

4.5.2.4.5.3 Fuel soak, phase II. Repeat 4.5.2.4.5.1, except fuel soak period shall be 7 days minimum with the flow control valve in the open position for 3 days, then in the closed position for 4 days.

4.5.2.4.5.4 Drying cycle, phase II. Repeat 4.5.2.4.5.2, except the drying period shall be 30 hours.

4.5.2.5 Cap, chain, and spring test. The dust cap, chain and spring assembly shall be subjected to a pull test of not less than 70 pounds. Nonconformance to 3.4.10.2 shall constitute failure of this test.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection.

4.6.1.1 Unit of product. The unit of product shall be a quantity of nozzles preserved and packaged in a single container.

4.6.1.2 Sampling. Sampling shall be in accordance with MIL-STD-105. Sample size shall be determined by using MIL-STD-105, table I and table IIa. A lot shall be accepted when zero defects are found and rejected when one or more defects are found.

4.7 Inspection comparison. The Government may select nozzles at any time during the contract production period and subject these nozzles to the examination specified in 4.5.1 and the tests specified in 4.5.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected first article nozzles. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of any nozzle to meet any requirement specified herein, during, and as a result of the examination and tests specified in 4.7, shall be cause for rejection of the nozzle and shall be cause for refusal by the Government to continue acceptance of production nozzles until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on nozzles produced and accepted under the contract. Any deficiencies found as a result of inspection comparison will be considered prim facie evidence that all nozzles

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accepted prior to completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.6.1.3 Examination. Samples selected in accordance with 4.6.1.2 shall be examined for the following major defects:

120. Nozzles not preserved as specified for level A or C (see 5.1.1 and 5.1.2).
121. Shipping containers not as specified for levels A and C (see 5.2.1 and 5.2.2).
122. Shipping container weight and size exceeding specified limitations for levels A and C (see 5.2.1 and 5.2.2).
123. Strapping not as specified for level A (see 5.2.1).
124. Marking incorrect, missing, or illegible (see 5.3).

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C, as specified (see 6.2).

5.1.1 Level A. Each nozzle shall be preserved in accordance with the method III requirements of MIL-P-116, utilizing a fiberboard carton as the unit container; the carton shall conform with PPP-B-636, class weather resistant, grade W6c. If necessary, cushioning material conforming to PPP-C-1752 or PPP-C-1842 shall be used.

5.1.2 Level C. Each nozzle shall be preserved as specified for level A, except the boxes shall be class domestic.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A. Nozzles, preserved as specified in 5.1 shall be packed in boxes conforming with PPP-B-601, overseas type, style optional. The gross weight of the shipping container and contents shall not exceed 150 pounds. Boxes shall be strapped in accordance with the appendix to the box specification except strapping shall be flat steel and shall be finish B.

5.2.2 Level C. Nozzles, preserved as specified in 5.1, shall be packed as specified for level A, except the boxes shall be domestic type and the strapping may be finish A.

5.3 Marking. Marking of both unit and shipping containers shall be in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The nozzle covered by this specification is intended for use in a low temperature (-60 °F) environment with power-operated dispensing devices for open-system transfer of aviation, motor and jet fuels from storage

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tanks, tank trucks, service station pumps, or other bulk containers to 5-gallon cans, 55-gallon drums, aircraft, motor vehicles, and motor vehicle tanks.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number, and date of the specification.
b. Nozzle size required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. When submission of first article nozzle(s) is not required (see 3.2).
e. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
f. When reducing bushings are required to be other than brass bar, brass casting, or aluminum (see 3.3.8).
g. When check valves are required (see 3.3.9).
h. When dust cap is not required to be attached (see 3.4.10.2).
i. Grounding cable required and length if other than as specified (see 3.4.10.3).
j. Level of preservation and packing required (see 5.1 and 5.2).

6.3 First article. When a first article inspection is required, the item(s) should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of the first article test results and disposition of the first articles. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each nozzle.

6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of nozzles.

6.6 Definitions.

6.6.1 Interchangeable parts. Interchangeable parts are defined as two or more like parts which possess such functional and physical characteristics as to be

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equivalent in performance and durability and which are capable of being exchanged one for the other without alteration of the parts themselves or of adjoining parts, except for adjustment, and without selection for fit or performance.

6.7 Subject term (key word) listing.

5-gallon drums
55-gallon drums
Arctic
Carrying handle
Cast aluminum
Check valve
Devices, dispensing
Fuel, aviation
Fuel, motor
Fuel hoses
Fuel, jet
Nozzle
Spout
Spouts
Springs, valves
Valve stem

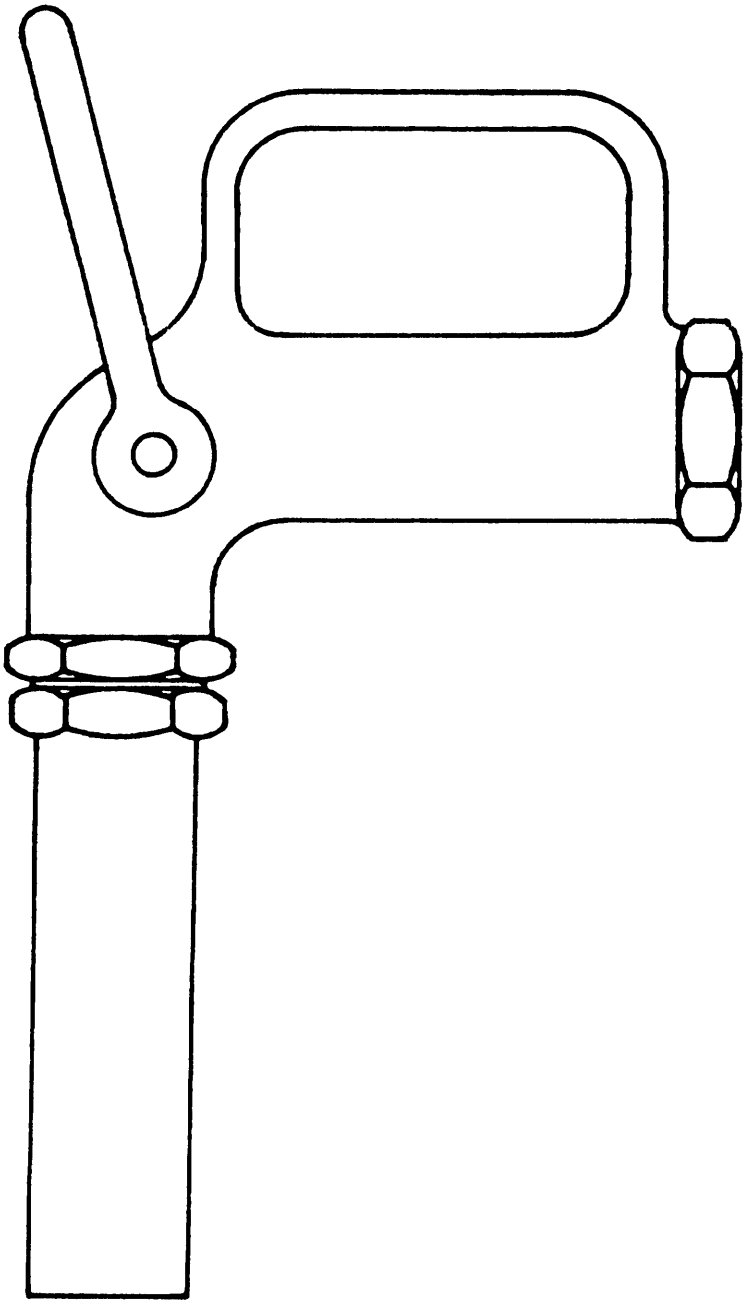
6.8 Figure 1. Figure 1 is for information purposes only.

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Army - ME

Preparing activity:
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NOTES:

1. NOZZLE CONFIGURATION IS ILLUSTRATIVE ONLY.

FIGURE 1. Nonautomatic shutoff nozzle.

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INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
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3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-N-53098 (ME)	90/02/07

3. DOCUMENT TITLE
Nozzles, Fuel and Oil Servicing, Nonautomatic Shutoff, Arctic Service

4. NATURE OF CHANGE (*Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.*)

5. REASON FOR RECOMMENDATION

6. SUBMITTER		
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON	e. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY	
a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial: (703) 664-5717 (2) AUTOVON: 354-5717
ADDRESS (Include Zip Code) US Army Belvoir RDE Center ATTN: STRBE-TSE Ft. Belvoir, VA 22060-5606	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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